LAB SHEET

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B. TECH CSE (CORE)

Database & Management System Lab (ENCS254)

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LAB SHEET - 1

Objective: To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

- **DDL Commands**

1. Create Database

```sql
CREATE DATABASE LAB_SHEET_1;
USE LAB_SHEET_1;
```

- **PART A – COLLEGE DATABASE**

1. Create Tables with Primary and Foreign Keys

```sql
CREATE TABLE STUDENT (
    USN VARCHAR(10) PRIMARY KEY,
    SName VARCHAR(50),
    Address VARCHAR(100),
    Phone VARCHAR(15),
    Gender VARCHAR(10)
);
```

2. Create SEMSEC Table

```sql
CREATE TABLE SEMSEC (
    SSID INT PRIMARY KEY,
    Sem INT,
    Sec VARCHAR(5)
);
```

3. Create CLASS Table

```sql
CREATE TABLE CLASS (
    USN VARCHAR(10),
    SSID INT,
    PRIMARY KEY (USN, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

4. Create SUBJECT Table

```sql
CREATE TABLE SUBJECT (
    Subcode VARCHAR(10) PRIMARY KEY,
    Title VARCHAR(50),
    Sem INT,
    Credits INT
);
```

5. Create IAMARKS Table

```sql
CREATE TABLE IAMARKS (
    USN VARCHAR(10),
    Subcode VARCHAR(10),
    SSID INT,
    Test1 INT,
    Test2 INT,
    Test3 INT,
    FinalIA INT,
    PRIMARY KEY (USN, Subcode, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

PART B – COMPANY DATABASE

1. Create DEPARTMENT Table

```sql
CREATE TABLE DEPARTMENT (
    Dno INT PRIMARY KEY,
    Dname VARCHAR(50),
    Mgrssn VARCHAR(15),
    Mgrstartdate DATE
);
```

2. Create EMPLOYEE Table

```sql
CREATE TABLE EMPLOYEE (
    SSN VARCHAR(15) PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    Sex VARCHAR(10),
    Salary DECIMAL(10,2) NOT NULL,
    Superssn VARCHAR(15),
    Dno INT,
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
```

3. Create DLocation Table

```sql
CREATE TABLE DLOCATION (
    Dno INT,
    Dloc VARCHAR(50),
    PRIMARY KEY (Dno, Dloc),
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
```

4. Create PROJECT Table

```sql
CREATE TABLE PROJECT (
    PNo INT PRIMARY KEY,
    PName VARCHAR(50),
    PLocation VARCHAR(50),
    DNo INT,
    FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
);
```

5. Create WORKS_ON Table

```sql
CREATE TABLE WORKS_ON (
    SSN VARCHAR(15),
    PNo INT,
    Hours DECIMAL(5,2),
    PRIMARY KEY (SSN, PNo),
    FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
    FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
);
```

- **PART C – CUSTOMER SALE DATABASE**

1. Create CUSTOMER Table

```sql
CREATE TABLE CUSTOMER (
    Cust_id INT PRIMARY KEY,
    Cust_name VARCHAR(50)
);
```

2. Create ITEM Table

```sql
CREATE TABLE ITEM (
    Item_id INT PRIMARY KEY,
    Item_name VARCHAR(50),
    Price DECIMAL(10,2) NOT NULL
);
```

3. Create SALE Table

```sql
CREATE TABLE SALE (
    Bill_no INT PRIMARY KEY,
    Bill_date DATE,
    Cust_id INT,
    Item_id INT,
    Qty_sold INT,
    FOREIGN KEY (Cust_id) REFERENCES CUSTOMER(Cust_id),
    FOREIGN KEY (Item_id) REFERENCES ITEM(Item_id)
);
```

- **ALTER OPERATIONS**

1. Add Email Column

```sql
ALTER TABLE STUDENT
ADD Email VARCHAR(100);
```

2. Rename SName to StudentName

```sql
ALTER TABLE STUDENT
CHANGE SName StudentName VARCHAR(50);
```

3. Drop DLOCATION Table

```sql
DROP TABLE DLOCATION;
```

- **DML Commands**

1. Inserted 10 records into CUSTOMER, ITEM and SALE tables.

```
INSERT INTO CUSTOMER VALUES (1,'Rohan');
INSERT INTO CUSTOMER VALUES (2,'Priya');
INSERT INTO CUSTOMER VALUES (3,'Aman');
INSERT INTO CUSTOMER VALUES (4,'Neha');
INSERT INTO CUSTOMER VALUES (5,'Karan');
INSERT INTO CUSTOMER VALUES (6,'Simran');
INSERT INTO CUSTOMER VALUES (7,'Arjun');
INSERT INTO CUSTOMER VALUES (8,'Meera');
INSERT INTO CUSTOMER VALUES (9,'Rahul');
INSERT INTO CUSTOMER VALUES (10,'Anjali');

INSERT INTO ITEM VALUES (101,'Laptop',50000);
INSERT INTO ITEM VALUES (102,'Mouse',500);
INSERT INTO ITEM VALUES (103,'Keyboard',1500);
INSERT INTO ITEM VALUES (104,'Bag',1200);
INSERT INTO ITEM VALUES (105,'Shoes',2500);
INSERT INTO ITEM VALUES (106,'Watch',3000);
INSERT INTO ITEM VALUES (107,'Book',300);
INSERT INTO ITEM VALUES (108,'Pen',50);
INSERT INTO ITEM VALUES (109,'Mobile',20000);
INSERT INTO ITEM VALUES (110,'Headphones',1800);

INSERT INTO SALE VALUES (1001,CURDATE(),1,101,1);
INSERT INTO SALE VALUES (1002,CURDATE(),2,102,2);
INSERT INTO SALE VALUES (1003,CURDATE(),3,105,1);
INSERT INTO SALE VALUES (1004,CURDATE(),4,109,1);
INSERT INTO SALE VALUES (1005,CURDATE(),5,103,3);
INSERT INTO SALE VALUES (1006,CURDATE(),6,104,2);
INSERT INTO SALE VALUES (1007,CURDATE(),7,106,1);
INSERT INTO SALE VALUES (1008,CURDATE(),8,107,5);
INSERT INTO SALE VALUES (1009,CURDATE(),9,110,2);
INSERT INTO SALE VALUES (1010,CURDATE(),5,101,1);
```

2. Bills of Current Date:

```
SELECT S.Bill_no, C.Cust_name, I.Item_name
FROM SALE S
JOIN CUSTOMER C ON S.Cust_id = C.Cust_id
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE S.Bill_date = CURDATE();
```

Output:

Result Grid | Filter Rows:

Bill_no	Cust_name	Item_name
1001	Rohan	Laptop
1002	Priya	Mouse
1003	Aman	Shoes
1004	Neha	Mobile
1005	Karan	Keyboard
1006	Simran	Bag
1007	Arjun	Watch
1008	Meera	Book
1009	Rahul	Headphones
1010	Karan	Laptop

3. Total Bill Amount:

```
SELECT S.Bill_no, (S.Qty_sold * I.Price) AS Total_Amount
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id;
```

Output:

Result Grid | Filter Rows:

Bill_no	Total_Amount
1001	50000.00
1002	1000.00
1003	2500.00
1004	20000.00
1005	4500.00
1006	2400.00
1007	3000.00
1008	1500.00
1009	3600.00
1010	50000.00

4. Customers who purchased products priced above 200.

```
SELECT DISTINCT C.Cust_name
FROM SALE S
JOIN CUSTOMER C ON S.Cust_id = C.Cust_id
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE I.Price > 200;
```

Output:

Cust_name
Rohan
Karan
Priya
Simran
Aman
Arjun
Meera
Neha
Rahul

5. Count number of products purchased by each customer.

```
SELECT C.Cust_name, COUNT(S.Item_id) AS Product_Count
FROM SALE S
JOIN CUSTOMER C ON S.Cust_id = C.Cust_id
GROUP BY C.Cust_name;
```

Output:

Result Grid | Filter Rows:

Cust_name	Product_Count
Rohan	1
Priya	1
Aman	1
Neha	1
Karan	2
Simran	1
Arjun	1
Meera	1
Rahul	1

6. Products purchased by customer with Cust_id = 5.

```
SELECT I.Item_name
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE S.Cust_id = 5;
```

Output:

Result Grid

Item_name
Keyboard
Laptop

- **Views Created**

1. Bill_Details View (shows bill_no, bill_date, cust_id, item_id, price, qty_sold, amount).

```
CREATE VIEW Bill_Details AS
SELECT S.Bill_no, S.Bill_date, S.Cust_id,
       S.Item_id, I.Price, S.Qty_sold,
       (I.Price * S.Qty_sold) AS Amount
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id;
```

```
SELECT * FROM Bill_Details;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Bill_no	Bill_date	Cust_id	Item_id	Price	Qty_sold	Amount
1001	2026-03-01	1	101	50000.00	1	50000.00
1002	2026-03-01	2	102	500.00	2	1000.00
1003	2026-03-01	3	105	2500.00	1	2500.00
1004	2026-03-01	4	109	20000.00	1	20000.00
1005	2026-03-01	5	103	1500.00	3	4500.00
1006	2026-03-01	6	104	1200.00	2	2400.00
1007	2026-03-01	7	106	3000.00	1	3000.00
1008	2026-03-01	8	107	300.00	5	1500.00
1009	2026-03-01	9	110	1800.00	2	3600.00
1010	2026-03-01	5	101	50000.00	1	50000.00

2. Weekly_Sales View (shows daily total sales for last one week).

```
CREATE VIEW Weekly_Sales AS
SELECT Bill_date, SUM(Qty_sold * Price) AS Total_Sales
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE Bill_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY Bill_date;
```

```
SELECT * FROM Weekly_Sales;
```

Output:

Result Grid | Filter Rows:

Bill_date	Total_Sales
2026-03-01	138500.00